Haoyang Jiang

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Education

- **Ph.D. in Data Science** *Expected Graduation: 2027*
  College of William & Mary, Williamsburg, VA
- **M.S. in Astrometry and Celestial Mechanics** *Sept. 2020 – Jun. 2023*
  University of Science and Technology of China, Hefei, China
- **B.S. in Geophysics (Minor in Computer Science)** *Sept. 2016 – Jun. 2020*
  China University of Mining and Technology, Beijing, China

Research Projects

- **Physics-Guided Modeling and Simulation** *[GitHub]*: Developing structure-aware graph operators and physics-guided updates for directional transport, stable spatiotemporal prediction, and long-horizon simulation of networked physical systems.
- **Neural Operators for Boundary Value Problems**: **FIE-NO** *[Paper]* uses Fredholm integral equations to learn solution operators under complex and varying boundary conditions.
- **Inverse Modeling with Generative Priors** *[GitHub]*: Studying full-waveform inversion with pretrained generative priors and inference-time optimization for reconstruction under new physical configurations.
- **Foundation Models for Forecasting and Data Assimilation**: Adapting pretrained spatiotemporal models for observation-guided long-horizon forecasting under sparse data and distribution shift.
- **Second-Order Optimal-Transport Flow Matching** *[GitHub]*: Developing a phase-space generative framework that couples position, velocity, and acceleration through an optimal-transport variational formulation.
- **Open-System and Boundary-Aware Learning** *[GitHub]*: Learning latent exterior influence through boundary proxies and physics-based refinement to stabilize prediction in partially observed physical domains.
- **Space Systems and Uncertainty Modeling**: Modeled satellite trajectories and studied orbit-error propagation under atmospheric-density uncertainty using machine-learning and numerical methods.

Publications

1. Jiang H. Y., Wang Z., Gao S., Zhang Y. J., Zhu X., He Y. *Physics-Refined Spatiotemporal Forecasting on Open-Boundary Hydrologic Graphs*. In **IEEE ICDM**, 2026.
2. Jiang H. Y., Qu B., Zhu X., Tan J., He Y. *Boundary-Consistent Graph Neural Networks for Topological Flux Prediction*. **Transactions on Machine Learning Research (TMLR)**, 2026.
3. Jiang H. Y., Qu Y. Z. *Fredholm Integral Equations Neural Operator (FIE-NO) for Boundary Value Problems.* Accepted by **Machine Learning: Engineering**, 2026.
4. Jiang H. Y., Wang J. D., He Y., et al. *Topology-aware Neural Flux Prediction Guided by Physics*. In **ICML**, 2025.
5. Jiang H. Y., Zhang M. J., et al. *Orbital error propagation considering atmospheric density uncertainty.* **Advances in Space Research**, 71(6), 2566–2574, 2023.

Honors and Awards

- **Outstanding Graduate Student Award**, University of Science and Technology of China, 2022, 2023
- **First-Class Scholarship**, China University of Mining and Technology (Beijing), 2018, 2019

Conferences for Invited Talks

- *Topology-Aware Graph Neural Networks by Physics*. KGML Workshop: Leading the New Paradigm of AI for Science, University of Michigan, Ann Arbor, Aug. 2025.
- *Satellite trajectory modeling using machine learning methods.* International Symposium on Space Flight Dynamics (ISSFD), Beijing, Aug. 2022.
- *Orbit error propagation considering atmospheric density uncertainty.* Youth Innovation Promotion Association of the Chinese Academy of Sciences, Nanjing, Dec. 2021.

Professional Service

- **Reviewer**: MM (2025), ICDM (2025, 2026), AAAI (2026), ICLR (2026), KDD (2026), NeurIPS (2026)

Internship Experience

ORIGIN SPACE, Nanjing, China *Feb. 2022 – Jun. 2022*
Assistant Engineer Intern

- Applied machine-learning and numerical methods in a space-technology R&D setting, reproducing and benchmarking research algorithms on engineering data.
- Improved research prototypes through data preprocessing, parameter tuning, and runtime optimization for engineering deployment.

Skills

- **Scientific ML**: Physics-Guided ML, Neural Operators, Graph Neural Networks, Generative Models, Data Assimilation

- **Scientific Computing**: Numerical Algorithms, PDE/ODE Solvers, Inverse Problems, Optimization
- **Programming**: Python, MATLAB, and relevant ML libraries